ClipSAT : Egyptian Scholastic Test: EST II Mathematics Level 2 [ANSWER KEY]

Unit 4.01: Matrices, Determinants & 2D/3D Vectors

Q1. A vector mathbf{v} is represented by the directed line segment from point P1, 2 to point Q4, 6 . What are the components of vector mathbf{v} ?

[x] A: [3, 4]

[ ] B: [5, 8]

[ ] C: [-3, -4]

[ ] D: [4, 3]

Solution: The components of a vector mathbf{v} represented by a directed line segment from point Px_1, y_1 to point Qx_2, y_2 are given by mathbf{v} = [x_2 - x_1, y_2 - y_1] . In this case, P = 1, 2 and Q = 4, 6 . Therefore, the components of mathbf{v} are [4 - 1, 6 - 2] = [3, 4] .

Q2. A translation maps point A1, 2 to A'-4, 2 and point B3, 5 to B'-2, 5 . This transformation is represented by begin{pmatrix} x' y' end{pmatrix} = begin{pmatrix} x y end{pmatrix} + begin{pmatrix} a b end{pmatrix} . Find the values of a and b .

[x] A: a = -5, b = 0

[ ] B: a = 5, b = 0

[ ] C: a = 0, b = -5

[ ] D: a = 0, b = 5

Solution: Subtracting the original coordinates of A1, 2 from its image A'-4, 2 gives the translation vector: begin{pmatrix} a b end{pmatrix} = begin{pmatrix} -4 - 1 2 - 2 end{pmatrix} = begin{pmatrix} -5 0 end{pmatrix}. Checking with B3, 5 to B'-2, 5 confirms -2 - 3, 5 - 5 = -5, 0 . Thus, a = -5 and b = 0 .

Q3. A linear transformation in the 2D plane is represented by the matrix M = begin{pmatrix} a & b c & d end{pmatrix} . This transformation maps the point P1, 2 to P'-3, 7 and the point Q3, 1 to Q'5, 1 . Determine the values of a, b, c, and d , and then calculate the sum a+b+c+d .

[x] A: 9

[ ] B: 11

[ ] C: 7

[ ] D: 5

Solution: Let the linear transformation matrix be M = begin{pmatrix} a & b c & d end{pmatrix} . The transformation of a point mathbf{x} to mathbf{x'} is given by Mmathbf{x} = mathbf{x'} . For point P1, 2 mapped to P'4, 9 : begin{pmatrix} a & b c & d end{pmatrix} begin{pmatrix} 1 2 end{pmatrix} = begin{pmatrix} 4 9 end{pmatrix} This yields the equations: 1 a + 2b = 4 2 c + 2d = 9 For point Q3, 1 mapped to Q'7, 12 : begin{pmatrix} a & b c & d end{pmatrix} begin{pmatrix} 3 1 end{pmatrix} = begin{pmatrix} 7 12 end{pmatrix} This yields the equations: 3 3a + b = 7 4 3c + d = 12 We need to solve these two systems of linear equations for a, b, c, and d . Solving for a and b using equations 1 and 3: From equation 3, b = 7 - 3a . Substitute this into equation 1: a + 27 - 3a = 4 a + 14 - 6a = 4 -5a = 4 - 14 -5a = -10 a = 2 Now substitute the value of a back into the expression for b : b = 7 - 32 = 7 - 6 = 1 Solving for c and d using equations 2 and 4: From equation 4, d = 12 - 3c . Substitute this into equation 2: c + 212 - 3c = 9 c + 24 - 6c = 9 -5c = 9 - 24 -5c = -15 c = 3 Now substitute the value of c back into the expression for d : d = 12 - 33 = 12 - 9 = 3 So, the transformation matrix is M = begin{pmatrix} 2 & 1 3 & 3 end{pmatrix} . The sum of the elements of the transformation matrix is a + b + c + d = 2 + 1 + 3 + 3 = 9 . Common misconceptions might arise from errors in solving the systems of equations, such as sign errors or algebraic mistakes when isolating variables. For instance, incorrectly solving for b or d could lead to a wrong sum. Alternatively, a student might forget to sum all four elements and only sum the elements of one row or column.

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Q4. A matrix M = begin{pmatrix} k & 2 3 & 1 end{pmatrix} transforms a point P1, 2 to P'5, 5 . Find the value of k .

[x] A: 1
[ ] B: 2
[ ] C: 3
[ ] D: 4

Solution: The transformation of a point Px, y by a matrix M = begin{pmatrix} a & b c & d end{pmatrix} is given by MP . In this case, we have begin{pmatrix} k & 2 3 & 1 end{pmatrix} begin{pmatrix} 1 2 end{pmatrix} = begin{pmatrix} 5 5 end{pmatrix} . Performing the matrix multiplication, we get: k1 + 22 = 5 and 31 + 12 = 5 . The second equation, 3 + 2 = 5 , is consistent. For the first equation, we have k + 4 = 5 . Solving for k , we subtract 4 from both sides: k = 5 - 4 , which gives k = 1 . Therefore, the value of k is 1.

Q5. Consider the matrix A = begin{pmatrix} 2 & 0 0 & 3 end{pmatrix} . When the matrix A is applied to the vector mathbf{v} = begin{pmatrix} x y end{pmatrix} , the resulting vector is Amathbf{v} = begin{pmatrix} 6 9 end{pmatrix} . What is the original vector mathbf{v} ?

[x] A: begin{pmatrix} 3 3 end{pmatrix}
[ ] B: begin{pmatrix} 1 2 end{pmatrix}
[ ] C: begin{pmatrix} 6 3 end{pmatrix}
[ ] D: begin{pmatrix} 12 27 end{pmatrix}

Solution: The matrix multiplication is given by: begin{pmatrix} 2 & 0 0 & 3 end{pmatrix} begin{pmatrix} x y end{pmatrix} = begin{pmatrix} 2x 3y end{pmatrix} We are given that this result is begin{pmatrix} 6 9 end{pmatrix} . Therefore, we have the system of equations: 2x = 6 3y = 9 Solving for x and y : x = frac{6}{2} = 3 y = frac{9}{3} = 3 So, the original vector mathbf{v} is begin{pmatrix} 3 3 end{pmatrix} .

Q6. Consider a linear transformation T represented by the matrix A = begin{pmatrix} 2 & 1 1 & 3 end{pmatrix} . Let P = begin{pmatrix} x y end{pmatrix} be a point in the plane. The transformation maps P to P' . If the area of the region transformed by T from a unit square in the xy -plane is A , and the determinant of A is detA , find the value of detA if the area of the transformed region is 5 .

[x] A: 5
[ ] B: 2
[ ] C: 1
[ ] D: 10

Solution: The area of the region transformed by a linear transformation T represented by a matrix A from a unit square is given by the absolute value of the determinant of A , i.e., Area = |detA|. In this problem, we are given that the matrix representing the transformation is A = begin{pmatrix} 2 & 1 1 & 3 end{pmatrix} . The determinant of A is calculated as: detA = 2 times 3 - 1 times 1 = 6 - 1 = 5 . We are also given that the area of the transformed region is 5 . Thus, |detA| = 5. Since we calculated detA = 5 , which is positive, the condition is satisfied. Therefore, the value of detA is 5 .

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Q7. Consider the transformation grid below. If point A has coordinates 1, 2 , what are the coordinates of its image, A', after the transformation shown?

[x] A: -4, 2

[ ] B: -2, 1

[ ] C: 2, -4

[ ] D: 1, -2

Solution: The transformation grid shows a reflection across the line y = x . When a point x, y is reflected across the line y = x , its image has coordinates y, x . Given point A is 1, 2 , its image A' will have coordinates 2, 1 . However, the provided `transformation_grid` JSON shows an object at `[[1, 2], [4, 2], [2, 5]]` labeled A, B, C and its image at `[[-4, 2], [-1, 2], [-3, 5]]` labeled A', B', C'. This implies a translation, not a reflection. Let's analyze the transformation of point A. Original point A is at 1, 2 . Its image A' is at -4, 2 . To find the translation vector begin{pmatrix} Delta x Delta y end{pmatrix} , we subtract the original coordinates from the image coordinates: Delta x = -4 - 1 = -5 and Delta y = 2 - 2 = 0 . So the translation is by the vector begin{pmatrix} -5 0 end{pmatrix} . The question asks for the coordinates of A' given A. Based on the figure's data, A' is -4, 2 . The provided choices are: -4, 2, -2, 1, 2, -4, 1, -2. The first choice matches the figure.

Q8. If A=begin{bmatrix}-4&-30&0end{bmatrix} and B=begin{bmatrix}-4&1-3&-2end{bmatrix} , find A+B .

[x] A: begin{bmatrix}-8&-2-3&-2end{bmatrix}

[ ] B: begin{bmatrix}0&-43&2end{bmatrix}

[ ] C: begin{bmatrix}-8&-1-3&-2end{bmatrix}

[ ] D: begin{bmatrix}-8&-2-4&-2end{bmatrix}

Solution: Add entrywise: A+B=begin{bmatrix}-8&-2-3&-2end{bmatrix} .